RESEARCH AND VERIFICATION REPORT | NOVEMBER 5, 2026

Pixels that matter. Decisions with consequences.

CausalFill-Bench v0.5.0 extends the earlier symbolic benchmark with actual image observations, calibrated spatial recovery and costly sensing before an irreversible simulated action. The release keeps the original tasks and proof artifacts, adds two task configurations, and records the tests supporting each new claim.

Complete test gate	Visual stimulus QA	Scripted control matrix
268 tests	624 PNG observations	3,780 episodes

The concrete advance

The agent must obtain information through the public sensor interface. A pixel reader calibrates a camera from visible landmarks, interprets a scene, and completes a decision. A separate robotics configuration assigns time and consequences to sensing, probing and execution. The evaluator retains the true scene and checks what actually happened. [E1-E5]

A stronger question than reconstruction alone

A recovered component matters when changing it changes a later decision. The release therefore measures recovery, outcome, observation cost and paired interventions separately. Its finite decision certificates also accept successful actions that do not identify every hidden detail. One checked fixture permits guaranteed action even though exact world identity remains unobservable. [E6]

> **What these results support**
>
> A reproducible, controlled image-to-decision research prototype with audited scripted references. The measured results concern its declared finite scenes and software. General image understanding, learned control and physical transfer require their own experiments.

Concept and direction: Phillip A. Holland Jr. Implementation, research synthesis and review: AI-assisted. Original project source and generated stimuli: MIT. Internal independent checkers provide different constructions; they are not external laboratory replication.

01 | WHAT CHANGED

From the earlier roadmap to executable experiments

The earlier report proposed noisy calibrated sensors, conservative action conditions, equal-budget controller comparisons and rendered scenes. Version 0.5 implements a bounded first step and identifies the experiments still needed beyond it. [E0-E5]

Earlier boundary	v0.5 implementation	Remaining evidence
Numerical camera coordinates	RGB PNG observations; visible landmarks; affine camera recovery	Natural scenes, depth, perspective, lens effects
No image occlusion study	Opaque shutters; byte-identical hidden twins; useful alternative views	3D occlusion and realistic camera-motion planning
Exact symbolic geometry	Rendered lane evidence and explicit simulated execution outcomes	Rigid-body contact dynamics and real actuators
Observation necessity only	Costed sensing, conservative alternatives and intervention controls	Strong learned controllers under enforced isolation
Finite identity recovery	Overlapping noisy supports and action-sufficiency certificates	Statistical calibration and simulator refinement proofs

Fourteen configurations, nine conceptual domains

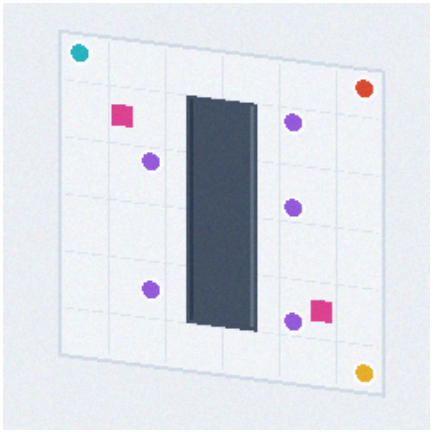
Language conditions, executable arithmetic, grid rules, procurement, robotics, finite-field state recovery, proof, diagnosis and visual perception share the typed tool protocol. The new visual family adds pose, spatial counting and dispatch recovery. The robotics_vision family joins the original robotics and barrier configurations. Earlier valid solutions remain accepted; new contracts are versioned separately.

The core still runs on Python 3.10 or newer without third-party runtime dependencies. Pillow is used for independent image QA and contact sheets; ReportLab builds this report. Those development dependencies do not become solver requirements.

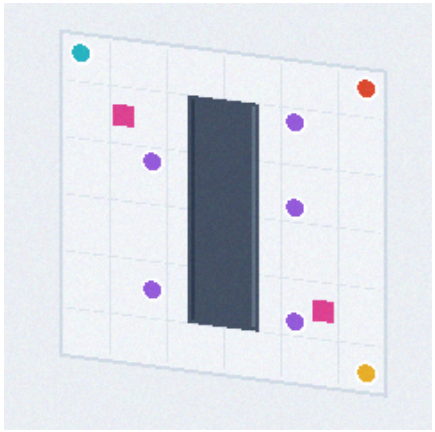
02 | THE PUBLIC PIXEL CONTRACT

Pixels are the public evidence

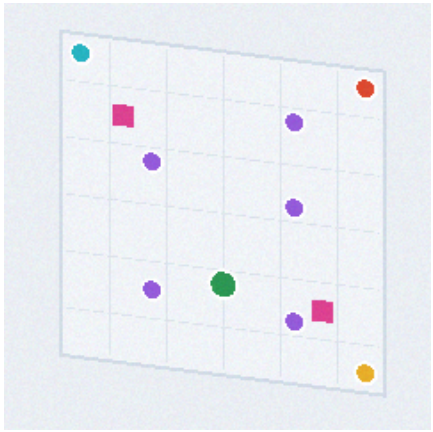
Each image payload contains PNG bytes, width, height, MIME type and SHA-256. The PNG contains only image header, compressed pixel data and end chunks. It carries no object coordinates, camera matrix, seed, segmentation channel or embedded answer. The contract separately declares colors, three landmark world points, sensor bounds and permitted tools. [E1-E2]



Occluded overview, twin 0



Occluded overview, twin 1



Useful inspection view

The two occluded overviews above are identical for the paired target intervention. Looking harder at that same image cannot recover the missing target location. The inspection view reveals the target while retaining the same declared camera and fixed noise realization. The image hash is included in the equivalence check; compression length is not ignored.

Calibrate before interpreting west

The affine map can rotate, reflect, translate and shear the scene. World-west means smaller world x after calibration, and need not mean image-left. Transfer cases include fractional pixel shifts, which produce measurable raster quantization error. The point estimate is checked against a declared 0.6 world-unit acceptance tolerance; it is not a probabilistic confidence interval.

The independent audit exhausts all 192 canonical transfer camera/target configurations. Its exact maximum centroid-derived coordinate error is 673/12012, about 0.05603 world units. This bound relies on the finite camera support, no clipping, and separated bounded RGB color supports. It does not calibrate a physical sensor. [E2]

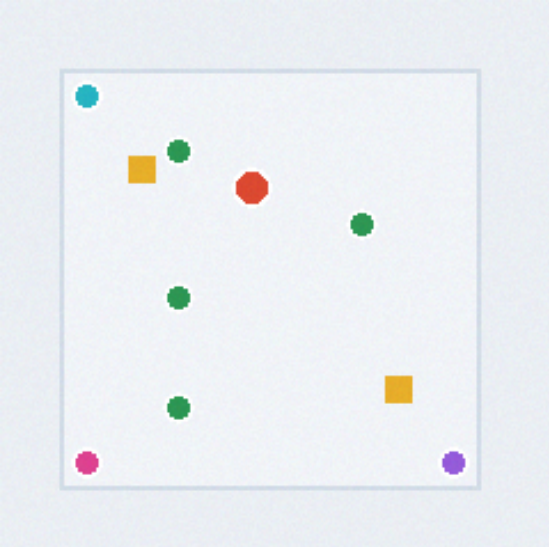
The standalone image client saves exactly the public PNG received over authenticated HTTP and returns its local path. A copied-client integration test checks the complete transport without importing evaluator code. The host must still enforce solver/evaluator isolation.

03 | CONTROLLED VISUAL REASONING

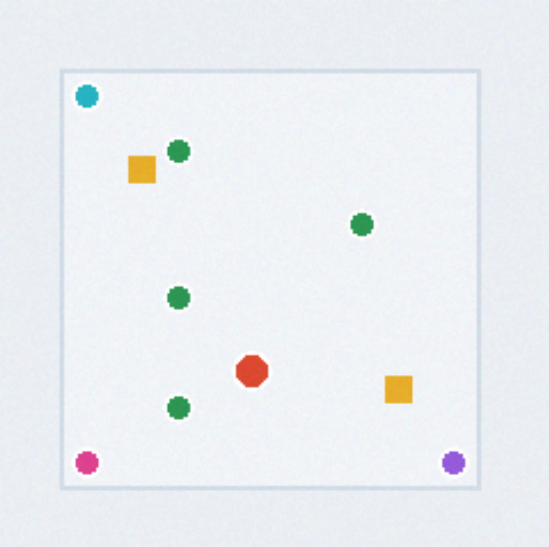
Recover a component, then use it

Requested completion	Independent acceptance condition
Target world position	Each coordinate lies within 0.6 world units of private geometry.
Token objects west of the target	Exact relational count computed from private object positions.
Dispatch branch	WEST when target x is below 48; EAST otherwise.

Strict success requires all three fields. Recovery and dispatch-rule correctness are reported separately, so a lucky correct branch does not hide a wrong pose. The visual task scores a submitted route label; mutable execution is tested in the robotics family. The scorer does not invoke the pixel reader or its calibration routine. Exact represented-decimal rational comparisons enforce the tolerance boundary. [E1-E2]



Same nuisance variables; first target state



Only target x changes; count and dispatch change

The generator changes one target coordinate between paired worlds. Colors, camera, noise, landmarks, distractors and other objects remain fixed. Randomized initial polarity avoids a fixed twin-to-answer convention. The development and transfer overviews conceal the entire changed target support.

The scene vocabulary is intentionally small: separated colored discs and rectangular distractors under a known legend. The task-aware detector is a solvability control for this contract. Its success should not be presented as a learned vision result or a measure of frontier-model difficulty.

04 | RENDERED ROBOTICS

Observation has a cost. Action has a consequence.

The new robotics configuration links rendered evidence to a finite operational task. The solver chooses sensing and handling actions through the same budgeted Session interface, and execution changes the simulated state. The outcome cannot be repaired by merely submitting the correct latent label afterward. [E3-E5]

An opaque roof hides which of two transport lanes is open. Moving to inspection exposes an actual PNG gate observation. Three visible fiducials establish the affine camera map. Camera changes, captures and probes consume simulated time, separately from synthetic tool cost.

A contact probe changes the surface through wear before returning a bounded friction interval. Gentle and firm contact therefore alter the later action conditions. Execution is terminal: a closed lane collides, insufficient traction slips, excessive fixture resistance stalls, and time or energy overruns prevent delivery.

Operation	Tool / time	Consequence
Move camera	2 / 1	Change station; use 1 energy.
Capture pixels	3 / 1	Fixed-pattern bounded noise; repeats are correlated.
Probe contact	4 / 2	Gentle adds 1 wear; firm adds 2. Reading refers to the changed surface.
Execute	2 / 5-18	A valid action commits once; delivery requires a successful route and mode within the public resource limits.

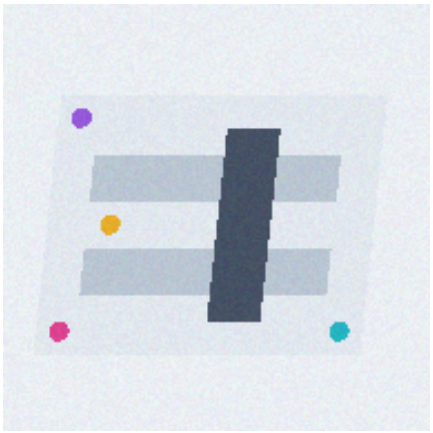
Conservative alternatives are legitimate

A robust action is accepted whenever it meets the same public completion, time and safety requirements. The scorer does not require a named sensor or maximum hidden-state recovery. A strict profile and a robust profile therefore test different decision problems.

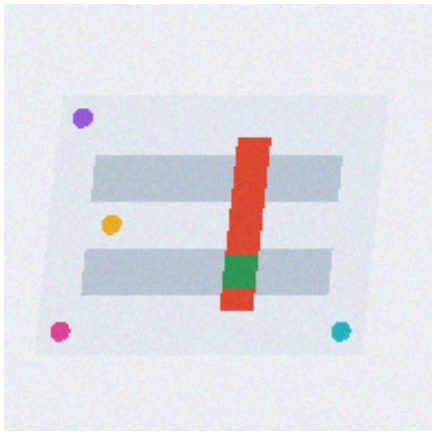
This is an authored finite control abstraction with raster observations. Its time, friction and damage rules are explicit simulator semantics. They are not identified parameters of a real robot, a physical safety certificate or a contact-rich 3D dynamics engine.

04A | TWO DISTINCT ROUTES TO A SUCCESSFUL DECISION

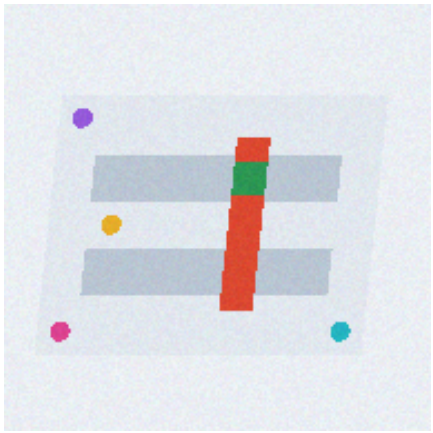
Observe the gate. Change the contact.



Roof hides the usable lane



Inspection reveals one opening



Gate twin reveals the other opening

These are actual 192 by 192 public camera observations. The overview is identical across gate twins; inspection distinguishes them. Friction twins have identical images in both views. The sensor contract therefore requires different evidence for route geometry and contact handling. [E3-E5]

Public contact profile	Largest initial friction	After one firm contact	Common grip succeeds?
Conditionable	0.65	0.41	Yes: both states are at or below 0.45.
Resistant	0.75	0.51	No: the dry state still exceeds 0.45.

Firm contact adds two wear units, reducing friction by 0.24 with a floor of 0.1. On conditionable surfaces, a controller can then use grip without interpreting the returned reading. On resistant surfaces, the same fixed-grip action may stall. These authored profiles test when changing the environment can substitute for identifying its earlier state.

The contact sensor refers to the surface after its own intervention. Repeated probing is neither a passive observation nor an independent sample. The task charges the extra time, energy and wear, and a failed execution remains terminal.

05 | ROBOT POLICY CONTROLS

Compare controllers on the same task contract

Seven scripted controls each receive the same tools and cap over 216 episodes. The default twelve-seed protocol covers every joint deadline/contact profile: two thirds strict and one third relaxed, with equal contact-profile counts. Tool budget: 30. Simulated deadlines: 10 strict, 26 relaxed.

Scripted policy	Success	Both twins	Tool cost	Obs. cost	Time
Budget-aware reference	216/216	108/108	9.0	4.7	12.0
Fixed with overview	216/216	108/108	15.0	10.0	10.0
Fixed minimal sensing	216/216	108/108	12.0	7.0	9.0
Query-free upper/roll	49/216	0/108	3.0	0.0	5.0
Robust slow detour	72/216	36/108	3.0	0.0	18.0
Common-mode conditioning	144/216	72/108	12.0	7.0	12.5
Firm then fixed grip	165/216	69/108	12.0	7.0	9.0

The minimal fixed sequence is a strong comparator: inspection plus a gentle probe suffices. The budget-aware reference skips sensing when the public deadline permits a slow detour. Its sensing schedule depends on that known contract; this is not a demonstrated observation-contingent sensing advantage.

Firm contact moves both initial regimes into the grip-safe range in the conditionable profile. The resistant profile preserves a case where this common-mode guarantee fails. The fixed-grip variant still succeeds on favorable individual states; complete-pair success exposes its missing guarantee.

Costs and time are per-episode means. Observation cost includes captures and probes; tool cost also includes the read, camera move and execution. A fixed sensing policy may use every returned observation to choose its terminal action. The source report retains collisions, slips, stalls, profile strata and sensor interventions. These are scripted control contrasts; transfer labels do not establish learned-policy generalization. [E3-E5]

06 | EXACT FINITE DECISION CERTIFICATES

Useful information can be less than complete identity

The new finite model represents possible worlds, terminal actions and sensor observation supports. A sensor may return several values in one world; the supports can overlap across worlds. The controller keeps every compatible world and must choose an action that succeeds in all of them. Each sensor is available once and has a positive integer cost. [E6]

Constructed fixture	Successful action	Full identification
Relevant lane only	1	4
Common robust action	0	4
Persistent blur ambiguity	No guarantee	No guarantee
Useful alternative noisy view	2	5
Conditional expensive probes	3	3
Action despite ambiguous identity	1	No guarantee

All terminal actions in this table cost zero, so the values compare observation costs. The general schema also permits positive action costs; action and identification totals are then different objectives and must not be subtracted as a universal information advantage.

A strict, finite adaptive advantage

In the conditional-probe fixture, a cheap selector followed by the relevant expensive probe costs 3 in the worst case. The best fixed sensor subset costs 4, even though its final action can use every returned observation. Complete alternatives are checked rather than comparing only a preferred reference against an obviously weak control.

The certificate verifies the minimum over legal stopping actions and sensor choices, including every needed nonempty observation branch. A zero-cost successful action permits pruning because no nonnegative alternative can improve it. The remaining sensor set shrinks even after an uninformative reading. An absent guarantee is represented explicitly.

07 | WHY ACTION COMPATIBILITY IS RICHER THAN A LABEL

An ambiguous world can still permit a reliable choice

Possible world	Camera support	Successful terminal action(s)
Left-only	L	Left
Shared	L or R	Left or right
Right-only	R	Right

After L, the world may be Left-only or Shared; Left succeeds in both. After R, it may be Shared or Right-only; Right succeeds in both. One observation therefore guarantees a successful action although neither result identifies the exact world. All four supported fixed-world observation paths are checked. [E6]

Avoid an incorrect shortcut in the abstraction

Sharing a successful action is not generally an equivalence relation. Left-only and Shared are compatible, and Shared and Right-only are compatible, but Left-only and Right-only share no successful terminal action. Taking a transitive closure would falsely merge the entire set. The implementation retains the actual action-success sets.

Three different obligations

The optimizer proposes a minimum and a selected action. A separate set-based checker validates every needed recurrence and alternative, binding the intended problem with SHA-256. A fixed-world traversal then replays the selected policy under every supported observation. The small-policy audit uses a further independent construction: enumerate complete syntactic decision trees and execute raw paths.

> **Independent finite-policy oracle**
>
> 1,377 exact optimum comparisons agreed with 101,898 complete syntactic policy evaluations over the declared tiny corpus. These counts measure finite checker coverage, not independent research tasks.

The proofs in this section are finite Python certificate checks. The historical Lean artifacts in the archive concern earlier formal tracks and retain their earlier versions. No new Lean proof or generic Python-to-physics equivalence is asserted.

08 | VISUAL VERIFICATION

Image QA checks meaning, visibility and failure modes

QA evidence	Count / scope
Actual observation PNGs	624: 336 visual + 288 robotics
Benchmark captures	576 from 288 task instances
Sensor diagnostic fixtures	48
Independent PNG decodes	624
Annotated contact sheets	39
Final visual review	39 sheets / 624 panels; AI inspection

Automated checks inspect RGB byte equivalence with Pillow, declared noise support, target visibility, fiducial calibration, pixel-derived completion, causal twin behavior and offscreen geometry. The independent raster oracle evaluates sampled pixel centers with exact rational arithmetic rather than calling the production shape classifier. [E1-E2]

The robotics gallery adds 288 captures through real Session tool calls. Those are image-QA sequences, not delivery trials. Its maximum observed opening-coordinate error is 0.2611 world units against the declared allowance of 1.0. The combined offline atlas includes all 624 observations with filters, zoom and separate evaluator annotations.

Partial visibility is a measured failure mode

A half-hidden disc has a biased visible-pixel centroid. The diagnostic fixtures record that bias and the visible fraction; they do not count the biased centroid as correct full-object recovery. Scored inspection views promise complete visible target support. Extending that promise to arbitrary occluded objects requires a different inference model.

The review changed the implementation

Development review found that integer-aligned scene centers made calibration artificially exact. Fractional pixel translations were added to the transfer distribution and verified at nonzero position error. Contact-sheet captions were shortened and checked against their card widths. Numeric input validation was hardened for extreme integers. These concrete findings are retained with their repaired follow-ups.

Noise levels outside the scored support, clipped objects and partial visibility are diagnostic controls. A generated image count is not a visual reasoning score. Contact sheets are evaluator artifacts containing annotations; only unannotated PNG observations belong in a scored solver environment.

09 | PAIRED IMAGE INTERVENTIONS

Do the delivered pixels cause the reference's success?

Delivered observation	Strict success	Complete pairs	Mean cost
Intact	72/72	36/36	4.0
Blank	0/72	0/36	4.0
Paired Swap	0/72	0/36	4.0
Forced Occlusion	24/72	12/36	4.0

The instrumented controls retain the same public task and costs while replacing the delivered pixels with a valid alternative PNG. Original hidden labels remain fixed. Intact-image success confirms solvability; blanking removes evidence; a paired swap supplies plausible evidence about the other causal world; forced occlusion removes the useful target only in the splits where the overview shutter exists. [E4]

Why these are useful negative controls

A pipeline could otherwise ignore images, recover labels from metadata, or claim perception success because its generator and answer function share state. These interventions test the actual sensor path used by the scripted reference. The paired swap changes its recovered answer and downstream dispatch; the failure is scored against the original world.

The test does not establish that no future model could exploit a public distribution shortcut. Prompt-only and fixed-action searches, independently authored mechanisms, and isolated learned-model runs remain necessary. All reported controls are source-bound development tests, with representative full traces and all episode records retained.

Robot interventions retain physical contact and timing rules while changing sensor evidence. On strict episodes, blanking forces the reference into a late detour; swapping gate-twin images produces collisions. False contact readings can cause stalls, while a conditioning policy that ignores the reading is unaffected. Relaxed reference episodes skip sensing and are explicitly unexposed controls. [E3]

> **Bandwidth is part of the experiment**
>
> A 240 by 240 RGB observation contains 172,800 uncompressed channel bytes before PNG compression. Image queries are not scalar measurements. The release records delivered bytes and dimensions separately from synthetic tool cost.

10 | RELEASE VERIFICATION

Stress evidence with honest units and provenance

Gate	Recorded result
Complete unit/integration discovery	268 passed; 0 failed; 0 errors
Full scripted suite	3,780 episodes; trace integrity checked
Robot control and intervention study	1,512 controls + 864 interventions
Independent combined audit	28,460 checks; 0 failures
Independent robot transition oracle	6,048 committed cases against exact rational conditions
Noisy-decision random fixtures	256 models; 512 certificates
Certificate forgery controls	3,072 rejected
Metamorphic comparisons	1,792
Independent finite policy comparisons	1,377
Original-release baseline	214 tests; 1,296 episodes; formal/protocol stress passed
New learned-model / physical trials	None in this release

The baseline ran from a fresh extraction of the original archive so concurrent development could not alter it. The final gates identify the new source version, commands, records and file hashes. A changed verifier or generator invalidates earlier source-bound conclusions until it is checked again. Development runs are retained separately from final runs. [E0-E7]

A replay scaling flaw found and repaired

Naive replay of synonymous observation labels could expand exponentially while visiting few distinct beliefs. The repaired replay combines equivalent fixed-world states, preserving exact path multiplicity and minimum/maximum costs. The regression represents over two billion label paths with 76 executed states. That is a compact exact accounting check, not billions of executed trials.

Hash chains detect modification relative to trusted evaluator records; they do not authenticate a malicious host. The local HTTP interface withholds grades and private state, but a shared filesystem is not enforced isolation. The benchmark should be deployed with a separate evaluator boundary for scored model experiments.

11 | TRANSFER AND THE 100D BOUNDARY

A disciplined next experiment

The release now supplies controlled pixel observations and simulated decisions. The next scientific question is whether a learned controller benefits from choosing evidence when nuisance changes, camera calibration is imperfect, and a robust action competes with further sensing. That requires a frozen model study rather than more generated seeds alone.

Stage	Concrete experiment	Advance criterion
Isolated model study	Freeze the task contract and compare active, fixed-sensing and robust controllers at matched caps.	A reproducible outcome/cost advantage on independently authored mechanisms.
Contact-rich simulation	Connect a physics simulator, separate privileged state from sensors, and audit transition correspondence.	Success under unseen object/contact configurations with measured abstraction errors.
Physical validation	Use a controlled laboratory task with measured sensors, actuator limits and intervention recovery.	Task completion and failure recovery supported by real trials under the lab protocol.

100D still means mathematical state dimension

The existing latent tasks recover operators over 8, 32 or 100 finite-field variables. Five probes at dimension 100 return 500 field values. The new image track does not turn these into 100 physical dimensions or prove improved robot perception. The useful shared idea is to account for what an observation reveals and whether that knowledge changes a successful decision.

Classical belief-space planning already links sensing and action [R1]. CLEVR motivates controlled visual diagnostics and distribution controls [R2]. ManiSkill's official interface distinguishes privileged states from sensor data [R3], and its physics simulator represents a later experimental layer with contact-rich dynamics [R4]. These are grounding sources, not evidence that this release matches their scope or results.

12 | EVIDENCE MAP AND COMMANDS

Reproduce, inspect and extend

```
python -m causalfill list

python -m causalfill demo --family visual --split transfer --difficulty 3

python -m causalfill demo --family robotics_vision --split transfer --difficulty 3

python -m unittest discover -s tests -v

python tools/stress_decision_v05.py --cases 256 --output private-runs/decisions.json

python tools/stress_vision_audit_v05.py --output private-runs/audit.json

python tools/render_visual_qa.py --seeds 8 --output private-runs/visual-qa
```

Use a new output directory for each run. Windows users may replace python with py. The commands run scripted development checks. An external model must receive only a public episode and the allowed tools, with its own inference budget and enforced isolation.

Evidence	Source-relative artifact
E0: prior baseline	reports/v05/legacy-baseline-summary.json
E1: generated pixel QA	reports/v05/visual-qa/manifest.json
Robot image QA / combined atlas	reports/v05/robotics-qa/manifest.json; reports/v05/qa-atlas.html
E2: independent audit	reports/v05/independent-vision-audit-final.json
E3: robot controls	reports/v05/robotics-controls/summary.json
E4: pixel interventions	reports/v05/visual-interventions/summary.json
E5: task contracts	docs/visual-perception.md; docs/robotics-vision.md
E6: noisy decision certificates	reports/v05/decision-stress-final.json; reports/v05/decision-certificates/
E7: consolidated release	reports/v05/release-verification.json

The archive includes per-file SHA-256 inventory, source, tests, configuration, generated stimuli and preserved historical proof evidence. The companion verification JSON records the exact release gates. These integrity records do not substitute for external replication.

13 | RESEARCH GROUNDING

Sources and claim boundaries

R1 | Integrated Task and Motion Planning in Belief Space

Acting, sensing and task planning under uncertainty are established research topics; use explicit beliefs and action consequences.

https://lis.csail.mit.edu/pubs/tlp/IJRRBelFinal.pdf

R2 | CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning

Controlled rendered data can expose shortcuts, compositional errors and attribute-distribution shifts. CausalFill includes no CLEVR data or code.

https://cs.stanford.edu/people/jcjohns/clevr/

R3 | ManiSkill observation modes

Separate privileged scene state from raw sensor observations in future simulator integration.

https://maniskill.readthedocs.io/en/latest/user_guide/concepts/observation.html

R4 | ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI

A contact-rich simulator is a materially different experimental layer from a dependency-free 2D raster fixture. No ManiSkill run is claimed.

https://arxiv.org/abs/2410.00425

Release claim

CausalFill-Bench v0.5.0 is an executable extension from symbolic recovery to controlled pixel evidence, costed simulated decisions, and exact finite decision-sufficiency auditing. Its references are inspectable and its negative controls are recorded. A broad breakthrough, state-of-the-art ranking, learned world-model result or physical robot improvement has not been measured by these tests.

The staged research direction is concrete: preserve observation integrity, accept successful alternatives, test what information changes action, measure the costs, and require separate evidence at each increase in realism. The full source and evidence make those next experiments reviewable and reproducible.